

Capture the Flag 6: Active or Inactive?

Goal: Make your Student Directory respond to student data by conditionally showing, hiding, and changing UI elements based on each student's state.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

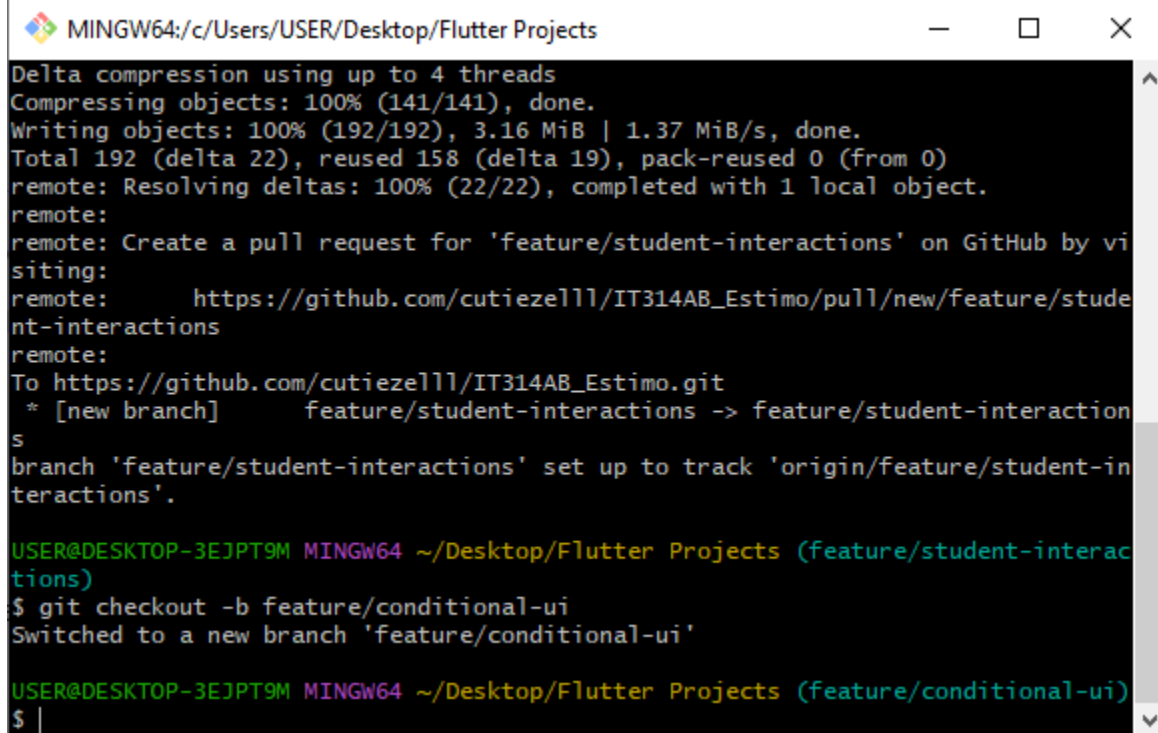
Objective: Create a new feature branch before adding conditional UI.

Create:

`feature/conditional-ui`

Screenshot:

- Terminal showing your current branch



```
MINGW64:/c/Users/USER/Desktop/Flutter Projects
Delta compression using up to 4 threads
Compressing objects: 100% (141/141), done.
Writing objects: 100% (192/192), 3.16 MiB | 1.37 MiB/s, done.
Total 192 (delta 22), reused 158 (delta 19), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (22/22), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/student-interactions' on GitHub by vi
siting:
remote:   https://github.com/cutiezell1/IT314AB_Estimo/pull/new/feature/stude
nt-interactions
remote:
To https://github.com/cutiezell1/IT314AB_Estimo.git
* [new branch]      feature/student-interactions -> feature/student-interac
s
branch 'feature/student-interactions' set up to track 'origin/feature/student-in
teractions'.

USER@DESKTOP-3EJPT9M MINGW64 ~/Desktop/Flutter Projects (feature/student-interac
tions)
$ git checkout -b feature/conditional-ui
Switched to a new branch 'feature/conditional-ui'

USER@DESKTOP-3EJPT9M MINGW64 ~/Desktop/Flutter Projects (feature/conditional-ui)
$ |
```

🚩 Flag 1 — Show Active or Inactive

Objective: Change the Student Card based on whether a student is active or inactive.

Add an **active** status to your student data.

Research how Flutter uses **conditional expressions** to display different widgets.

Example states:

- Active
- Inactive

Test

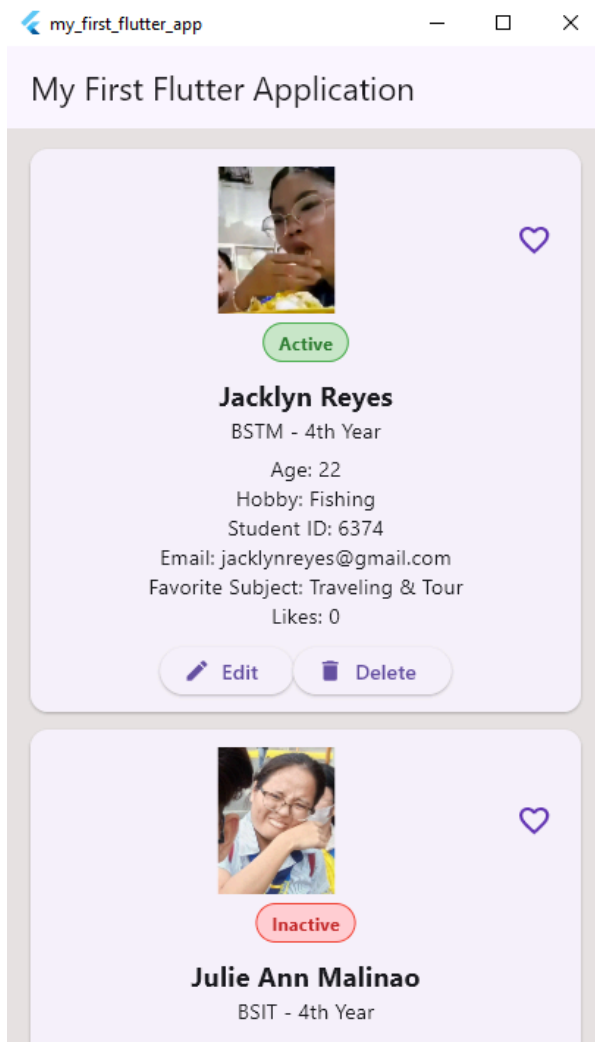
Active students should display an Active indicator.

Inactive students should display an Inactive indicator.

Screenshot:

- Active Student Card

- Inactive Student Card



- Conditional logic in your code

```
child: Text(
  student.active ? 'Active' : 'Inactive',
  style: TextStyle(
    fontSize: 12,
    fontWeight: FontWeight.bold,
    color: student.active ? Colors.green[800] : Colors.red[800],
  ), // TextStyle
), // Text
```

🚩 Flag 2 — Show or Hide

Objective: Display certain UI elements only when a condition is true.

Add a conditional element to the Student Card.

Examples:

- Favorite badge appears only when favorited
- Warning appears only when inactive
- Delete action appears only if active

Research ways to conditionally include or exclude widgets from the widget tree.

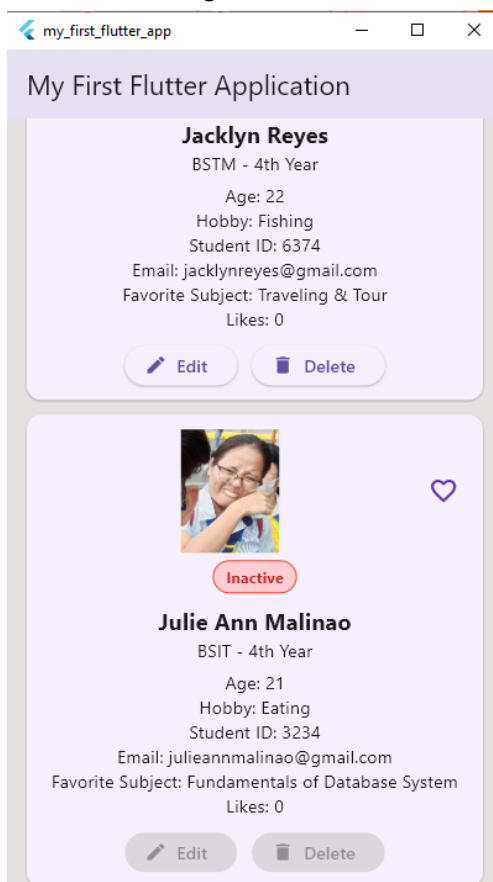
Test

When the condition is true, the element appears.

When the condition is false, the element disappears.

Screenshot:

- UI with the element visible
- UI with the element hidden
- Conditional widget code



🚩 Flag 3 — The Empty Directory

Objective: Handle situations where there are no students to display.

Modify your Student Directory so it detects when the student list is empty.

Create an **empty state**.

Examples:

- "No students found"
- Empty-state icon
- Button to add a student

Test

Delete all students in your application (not the code)

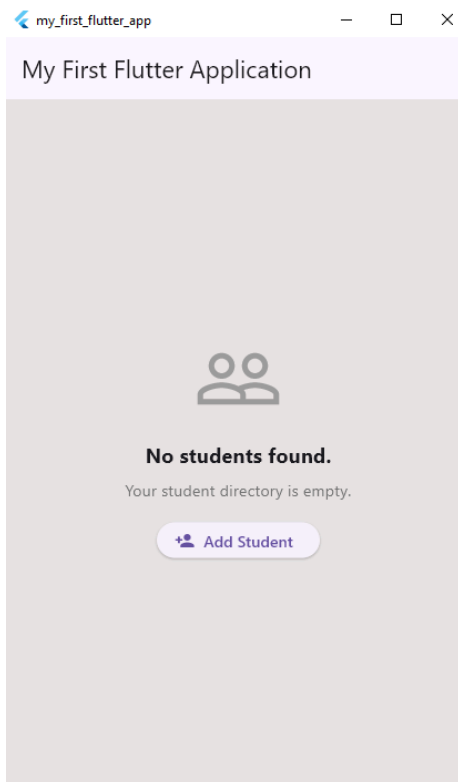
Instead of showing a blank screen, your application should display your empty state.

Screenshot:

- Empty student list

```
List<Student> students = [];
```

- Empty-state UI



- Condition checking for an empty list

```
Widget emptyState(BuildContext context) {  
  return Center(  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.center,  
      children: [  
        Icon(Icons.people_outline, size: 80, color: Colors.grey[500]),  
        const SizedBox(height: 16),  
        const Text(  
          'No students found.',  
          style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),  
        ), // Text  
        const SizedBox(height: 8),  
        Text(  
          'Student directory is empty.',  
          style: TextStyle(fontSize: 14, color: Colors.grey[600]),  
        ), // Text  
        const SizedBox(height: 20),  
        ElevatedButton.icon(  
          onPressed: () => addStudent(context),  
          icon: const Icon(Icons.person_add),  
          label: const Text('Add Student'),  
        ), // ElevatedButton.icon  
      ],  
    ),  
  );  
}
```

🚩 Flag 4 — Loading the Directory

Objective: Display different UI while student data is being loaded.

Create a loading state for your Student Directory.

Research how Flutter can conditionally display a loading indicator.

Examples:

- `CircularProgressIndicator`
- Loading message
- Loading placeholder

Test

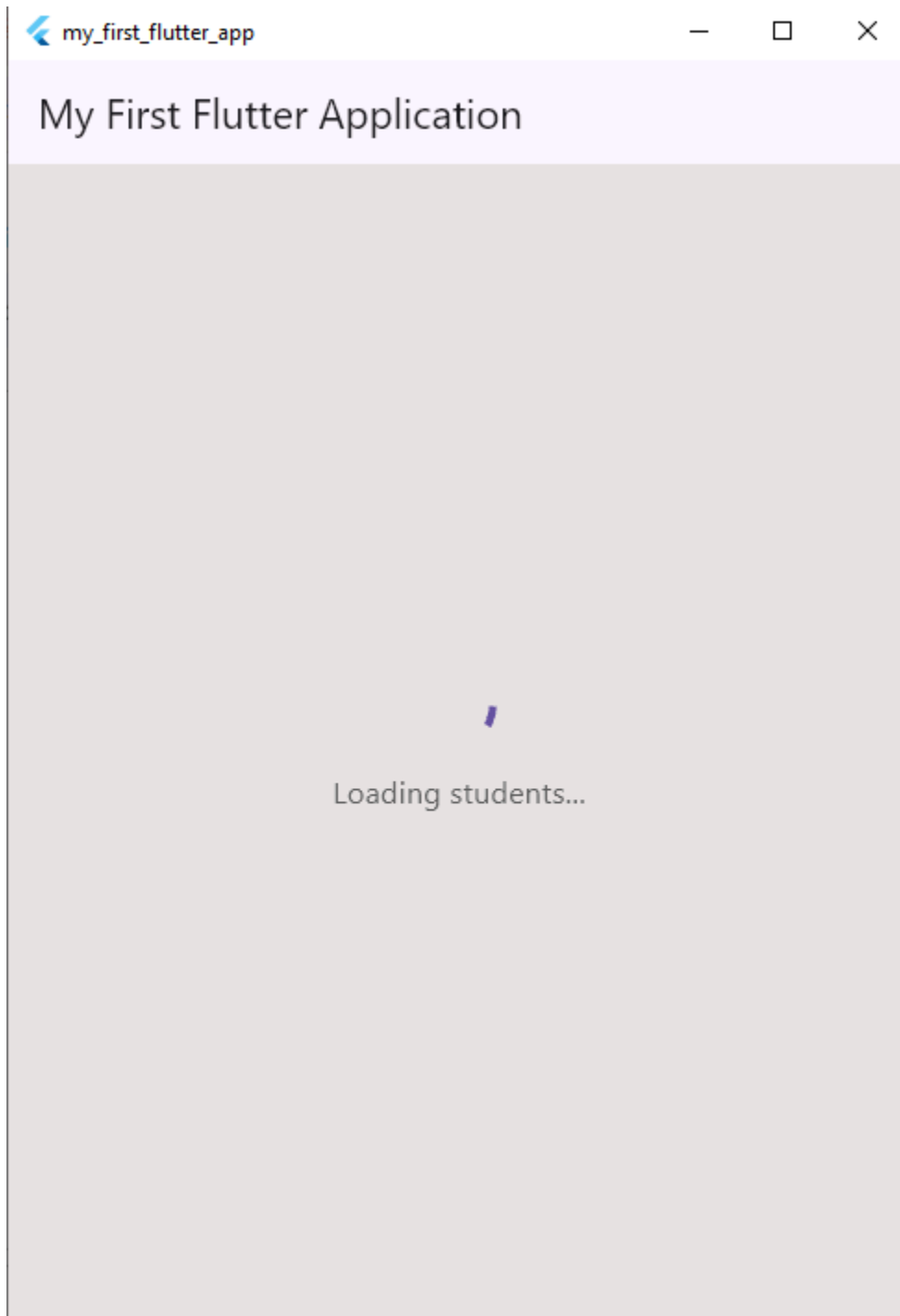
When the application is in a loading state, the student list should not be displayed as the main content.

Once loading is complete, the Student Directory should appear.

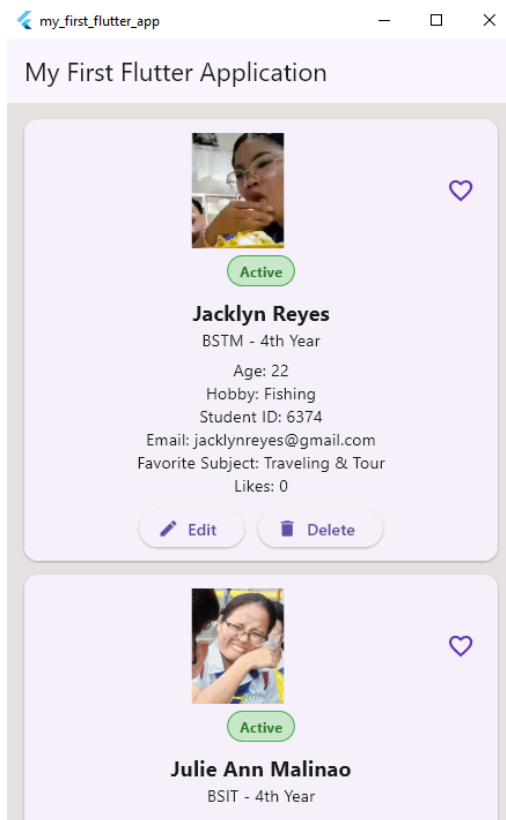
Screenshot:

P.S (Miss, nag ka problem ko sa emulator since 4RAM rajud ako lappy nya sa physical na device which is ako phone kay nag ka problem pud huhueness)

- Loading state



- Loaded state



- Conditional loading logic

```
bool isLoading = true;

@override
void initState() {
  super.initState();
  loadStudents();
}

Future<void> loadStudents() async {
  await Future.delayed(const Duration(seconds: 5));
}
```

🚩 Flag 5 — Multiple States

Objective: Make your Student Directory handle multiple UI states correctly.

Your application should now be able to display:

1. Loading
2. Students available
3. No students available

Determine which state should be displayed based on your data.

Test

Verify each state:

Loading

→ Loading UI

Students available

→ Student List

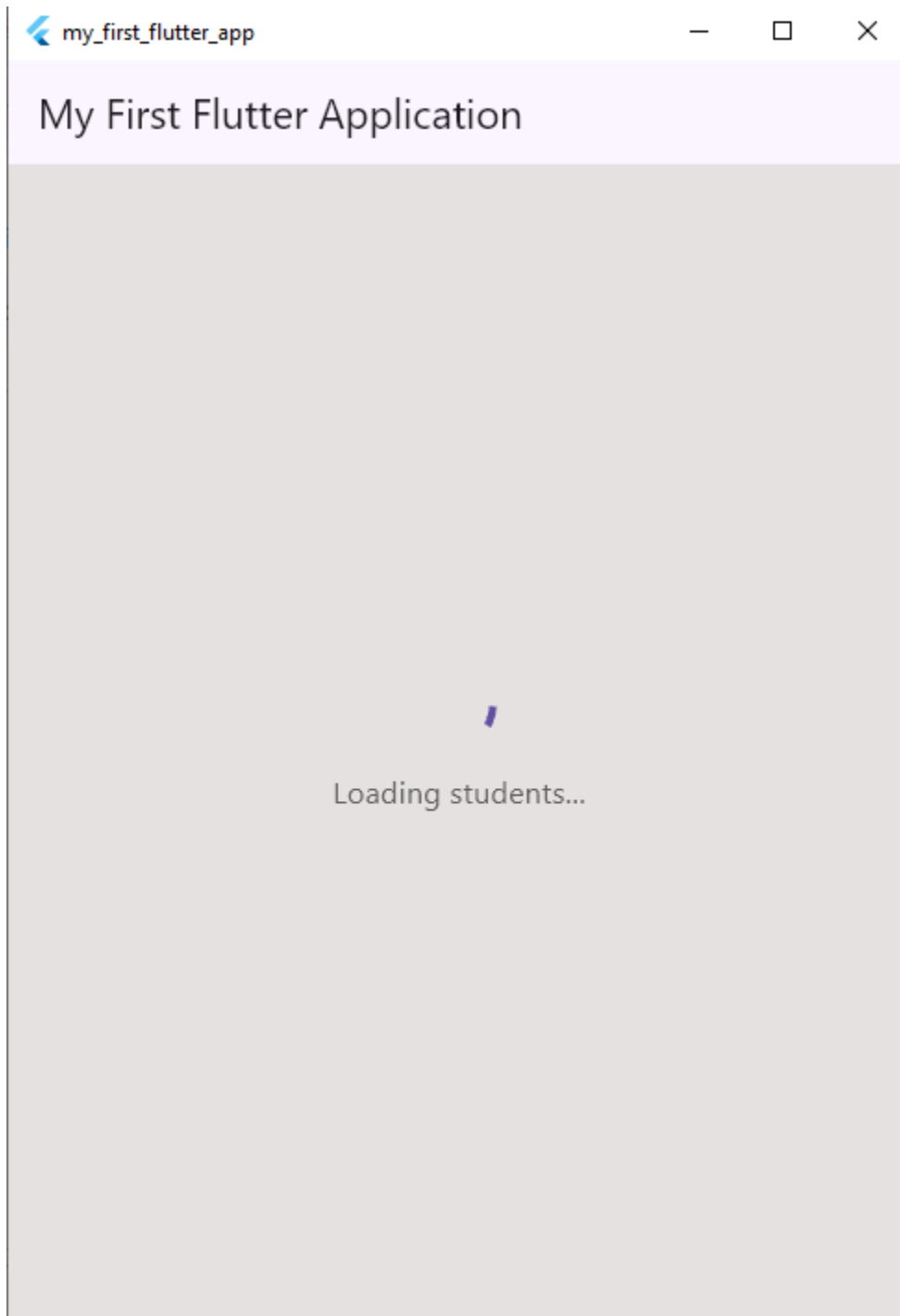
Empty list

→ Empty State

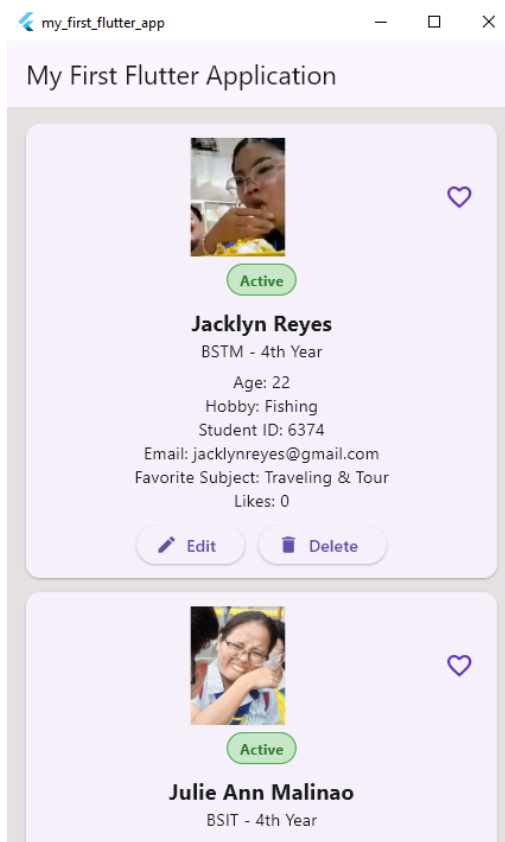
Only the appropriate UI should appear at a time.

Screenshot:

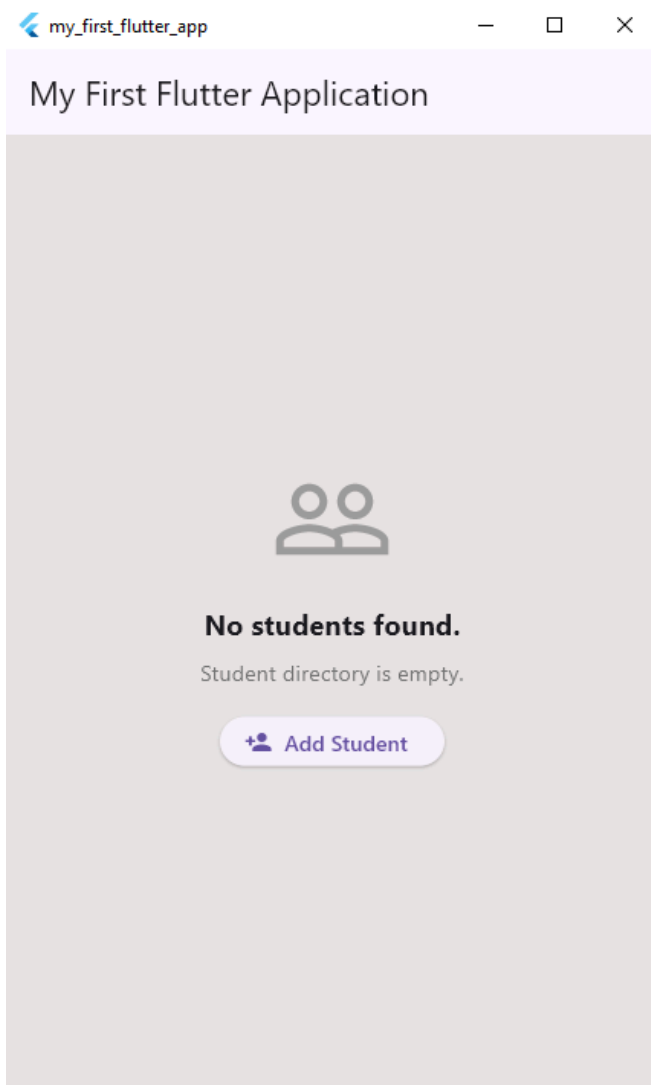
- Loading state



- Student list state



- Empty state



🚩 Flag 6 — States Work Together

Objective: Make sure conditional UI does not break your existing interactions.

Test the combination of:

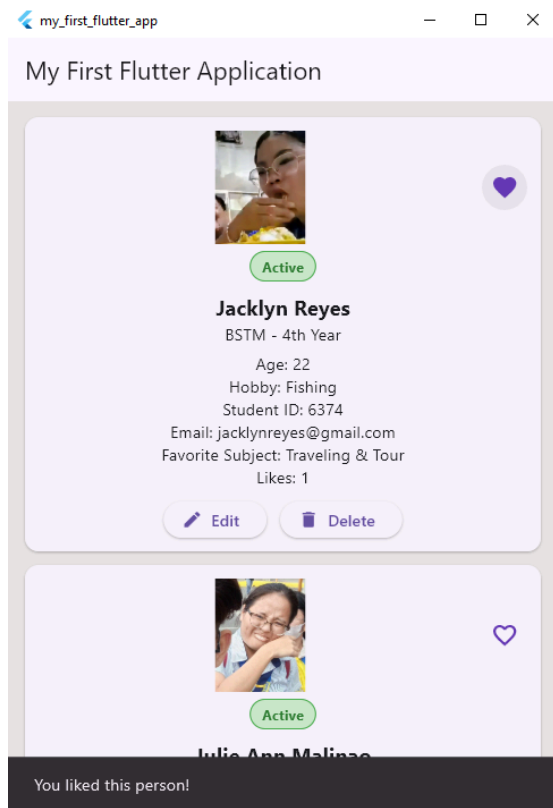
- Favorite
- Delete
- Edit
- Active/Inactive
- Show/Hide
- Empty state
- Loading state

Test Scenario

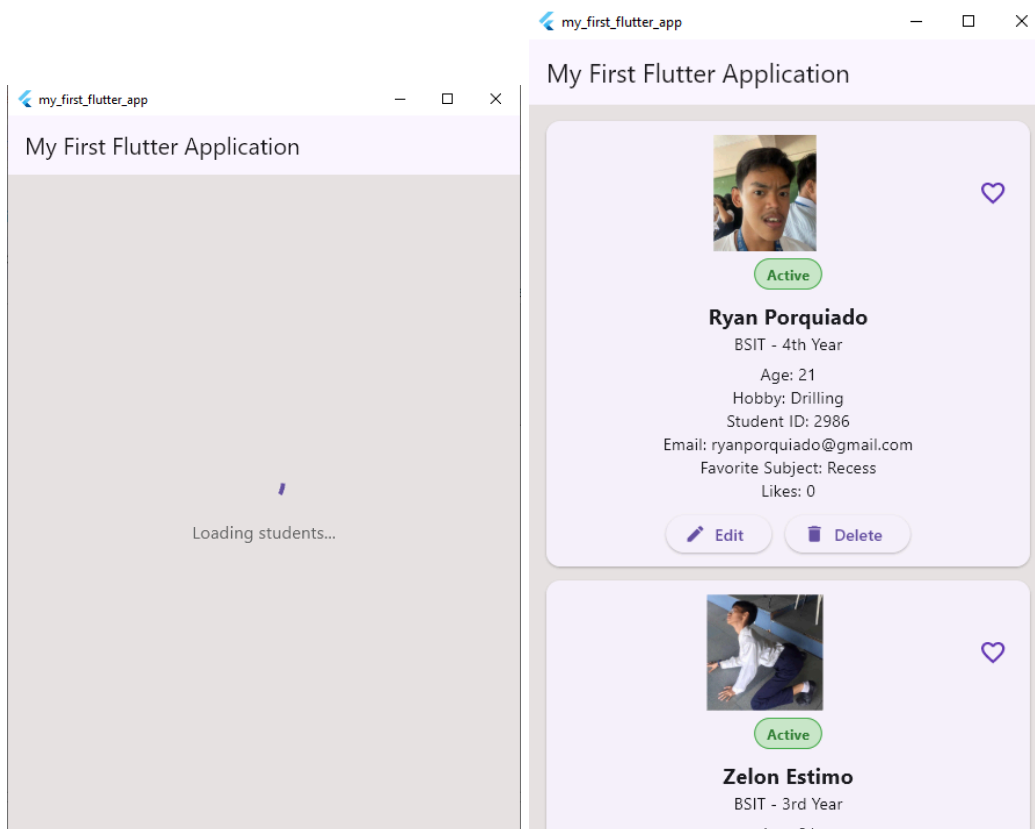
1. Favorite a student.
2. Change or display their Active/Inactive state.
3. Delete students until the directory becomes empty.
4. Verify the Empty State appears.
5. Reload or restore students.
6. Verify the Student Directory appears again.

The application should continue functioning correctly as its state changes.

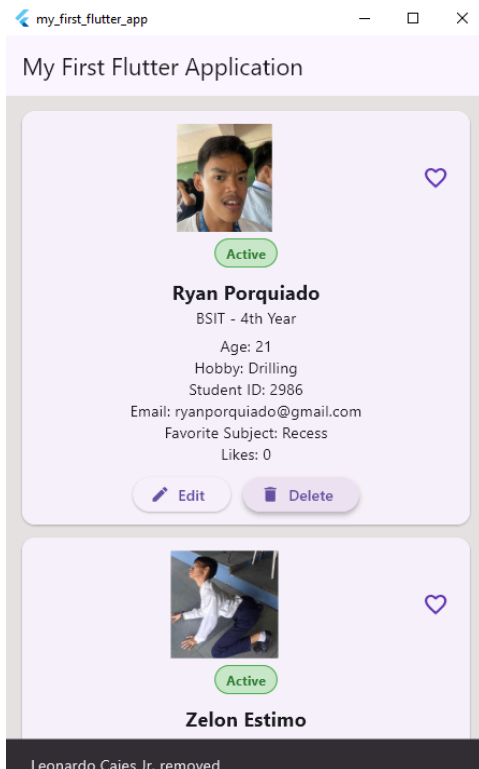
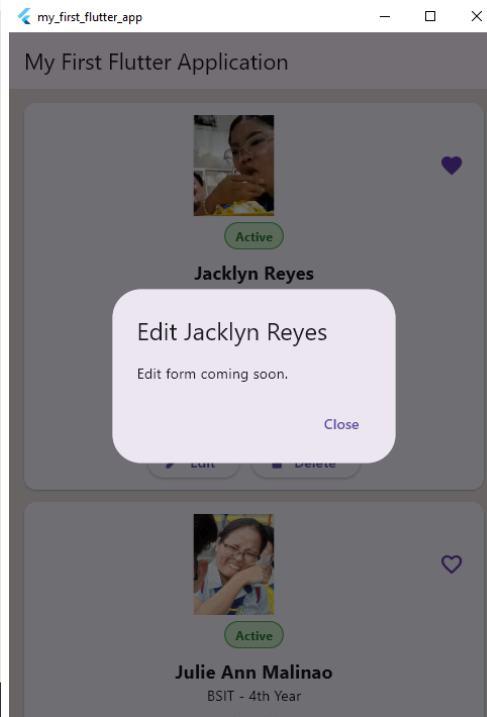
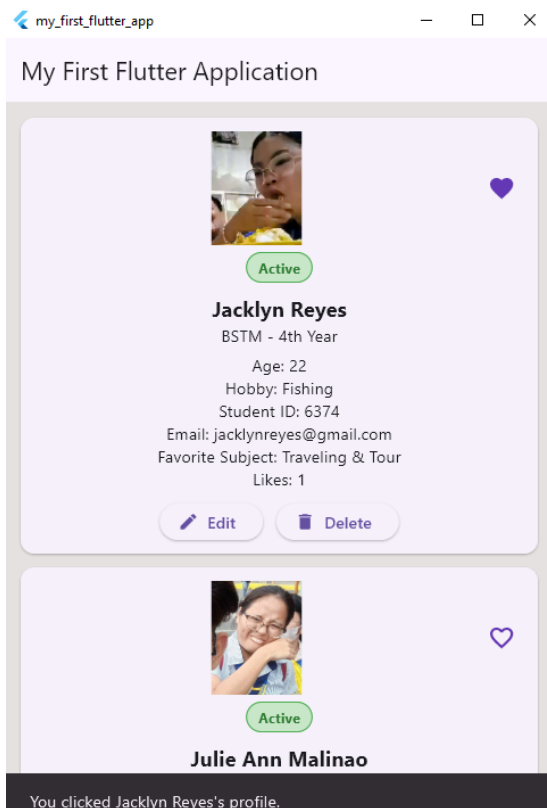
Screenshot:



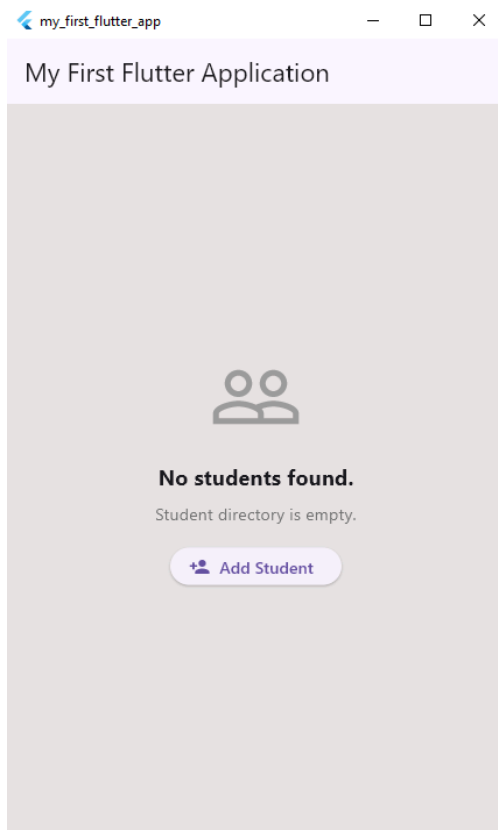
- Multiple conditional states working



- Interactive Student Card



- Empty State



🚩 Flag 7 — Save Your Progress

Objective: Commit and push your completed Conditional UI Student Directory.

Use a proper Git commit message.

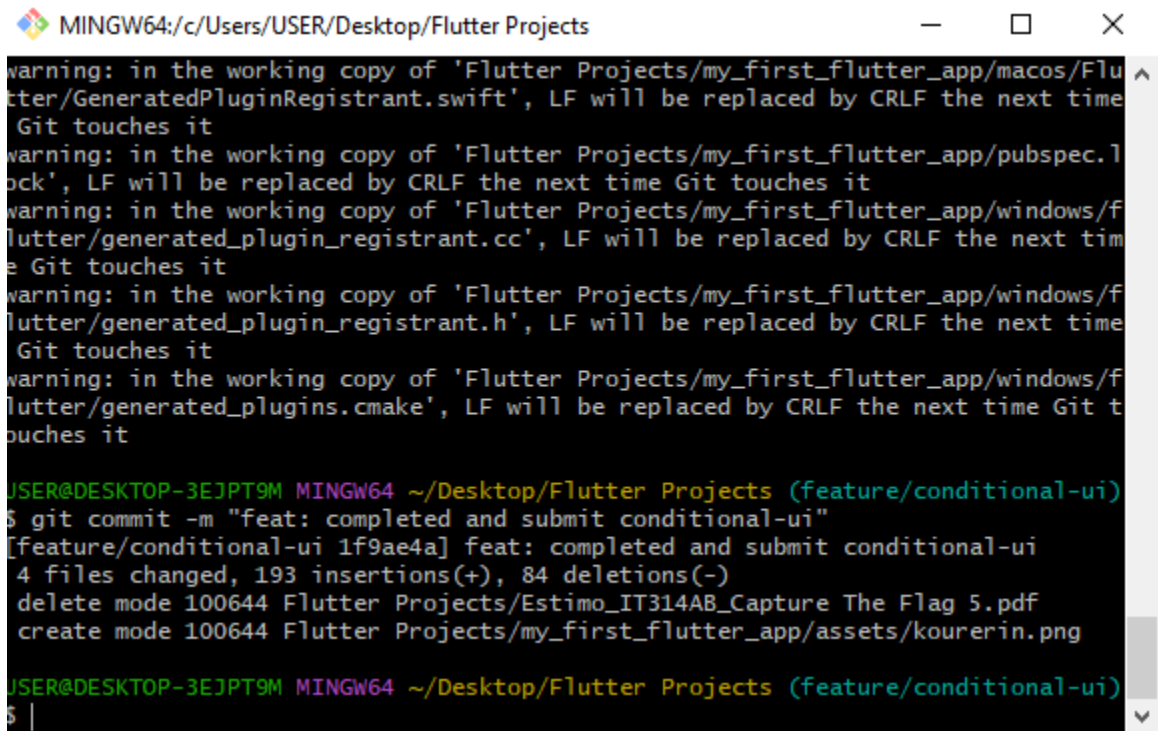
Branch:

`feature/conditional-ui`

Screenshot:

- Successful commit
- Current branch

- Terminal showing the commit

A screenshot of a Windows terminal window titled 'MINGW64:/c/Users/USER/Desktop/Flutter Projects'. The terminal shows several warning messages about line ending conversions (LF to CRLF) for various files in the Flutter project. After these warnings, the user enters the command 'git commit -m "feat: completed and submit conditional-ui"'. The terminal then displays the commit output, including the commit hash '1f9ae4a', the commit message, and a summary of file changes: '4 files changed, 193 insertions(+), 84 deletions(-)'. It also shows the deletion of a PDF file and the creation of a PNG asset file.

```
MINGW64:/c/Users/USER/Desktop/Flutter Projects
warning: in the working copy of 'Flutter Projects/my_first_flutter_app/macos/FlutterGeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'Flutter Projects/my_first_flutter_app/pubspec.lock', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Flutter Projects/my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'Flutter Projects/my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'Flutter Projects/my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git t
ouches it

USER@DESKTOP-3EJPT9M MINGW64 ~/Desktop/Flutter Projects (feature/conditional-ui)
$ git commit -m "feat: completed and submit conditional-ui"
[feature/conditional-ui 1f9ae4a] feat: completed and submit conditional-ui
4 files changed, 193 insertions(+), 84 deletions(-)
delete mode 100644 Flutter Projects/Estimo_IT314AB_Capture The Flag 5.pdf
create mode 100644 Flutter Projects/my_first_flutter_app/assets/kourerin.png

USER@DESKTOP-3EJPT9M MINGW64 ~/Desktop/Flutter Projects (feature/conditional-ui)
$ |
```